Numerical Analysis I

Quiz 8A Name:_____

Question 1.

Consider the function

$$f(x) = \frac{1}{x}, \qquad x > 0.$$

We want to approximate $f(x)$ using polynomial interpolation at the four nodes

$$x_0 = 1, \qquad x_1 = 2, \qquad x_2 = 3, \qquad x_3 = 4.$$

(a) Complete the following Newton divided difference table. The blank entries are to be computed.

x_i	$f[x_i]$	First differences	Second differences	Third differences
1		$-\frac{1}{2}$	$\frac{1}{6}$	
2		$-\frac{1}{6}$		
3		$-\frac{1}{12}$		
4				

(b) Using the divided differences from part (a), write the polynomial interpolant $p_3(x)$ of degree at most 3 in *Newton's nested form.*

(c) Use the polynomial interpolation error theorem to find an upper bound for

$$|f(2.5) - p_3(2.5)|.$$

Question 2.

1. What is the definition of the Chebyshev polynomials of the first kind $T_n(x)$?

2. Derive the three-term recurrence relation for the Chebyshev polynomials T_n.

3. We are given
$$T_3(x) = 4x^3 - 3x$$
and
$$T_4(x) = 8x^4 - 8x^2 + 1.$$
Use the recurrence relation from part (b) to compute $T_5(x)$.

4. Find all the zeros of $T_5(x)$.

5. Sketch the graph of $T_5(x)$ on $[-1, 1]$. Clearly indicate its zeros and its values at the points where it attains 1 or -1.

6. State (without proof) the minimax property of the Chebyshev polynomial T_5. In other words, explain what makes T_5 special among polynomials of degree 5 with the same leading coefficient.